

Assignment 4 Experiments

Due: March 20th, 2026

– Group 111 –

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Question 3: Experiments I

a) Size of the Encoding

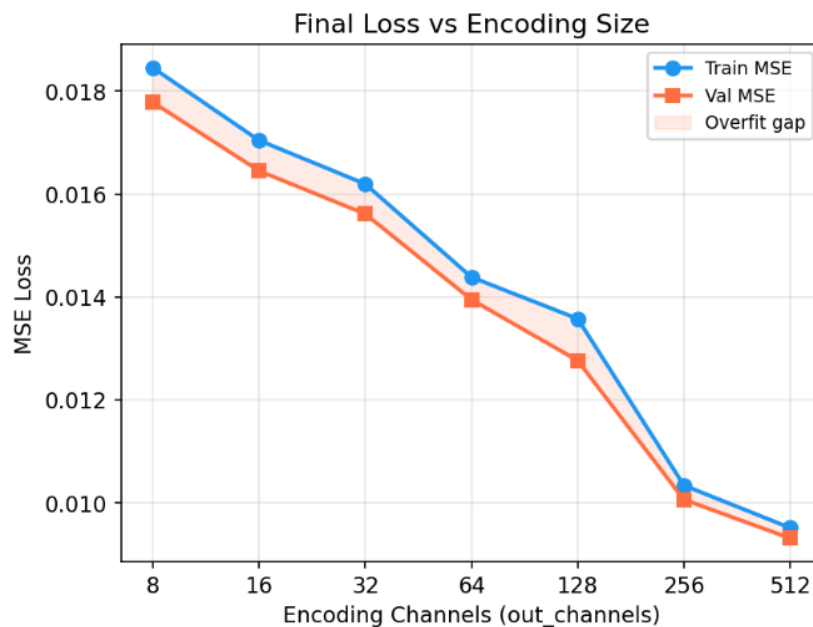


Figure 1: Loss (MSE) for different encoding sizes

Both train and val MSE decrease steadily as encoding size grows, with no plateau even at 512 channels. The model is still gaining from more capacity, so encoding size isn't the limiting factor here.

b) Number of Convolutional Layers

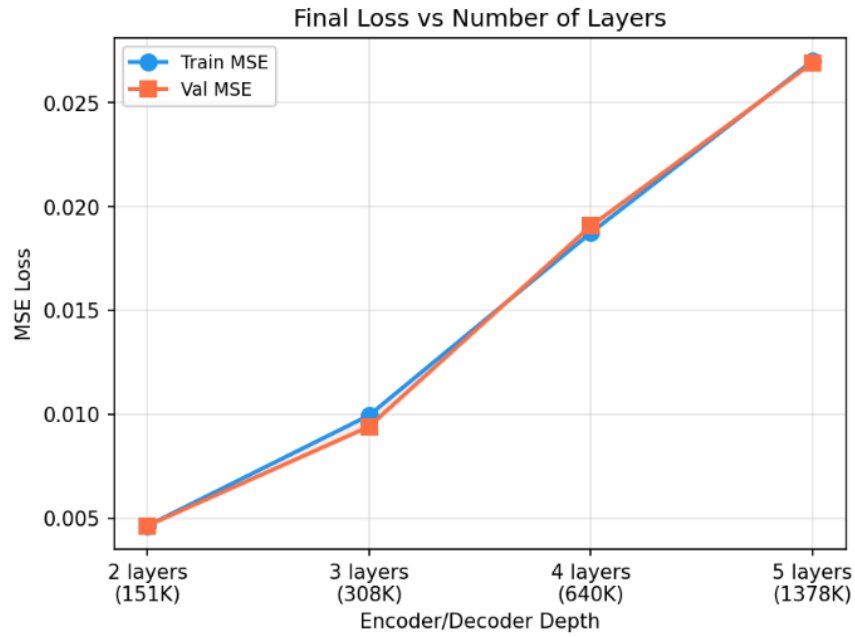


Figure 2: Loss (MSE) for different number of layers

The 2-layer model gets the best loss (~ 0.005) while 5 layers is the worst (~ 0.025). Each layer halves the spatial size, so by 5 layers, 32×32 images shrink down to 1×1 , and the decoder struggles to recover from that. For images this small, 2–3 layers is plenty.

c) Number of Epochs

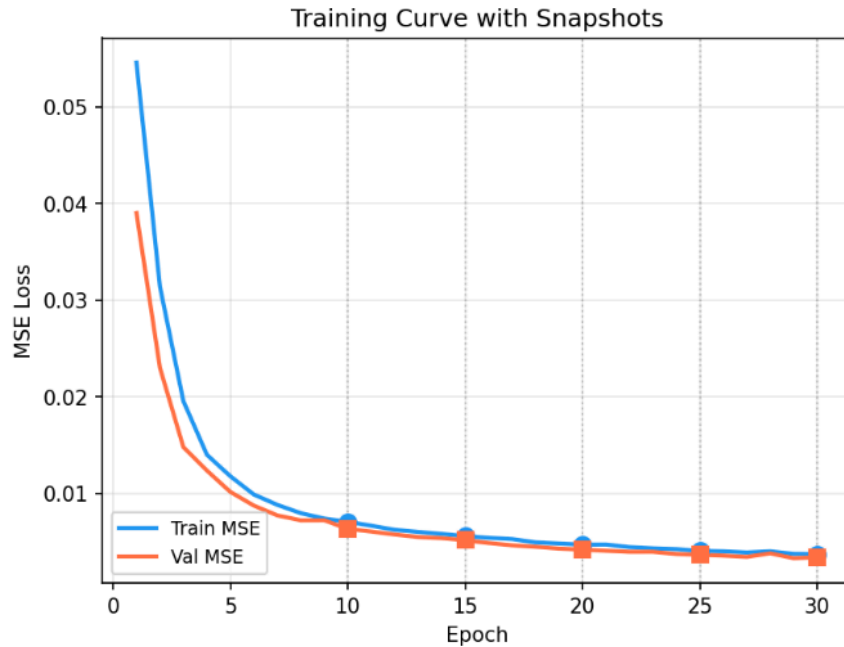


Figure 3: Loss (MSE) for different number of epochs

Most of the learning happens in the first ~10 epochs, with the curve dropping sharply from 0.055 to around 0.007. The model keeps improving after that but the gains get smaller and smaller. By epoch 20–25 the curve is basically flat, so training much beyond that may not be worth it.

d) Best Performing Model

Optimal parameters: 512 channel encoding, 2 convolutional layers, 25 epochs.

Results (MSE Loss):

Training Set: 0.00411

Validation Set: 0.00371

Test Set: 0.00354

Question 4: Experiments II

a) Pre-processing

Note that for normalized pre-processing, the output layer of the decoder is using tanh, whereas “unnormalized” ($[0, 1]$) is using Sigmoid.

Baseline (no preprocessing):

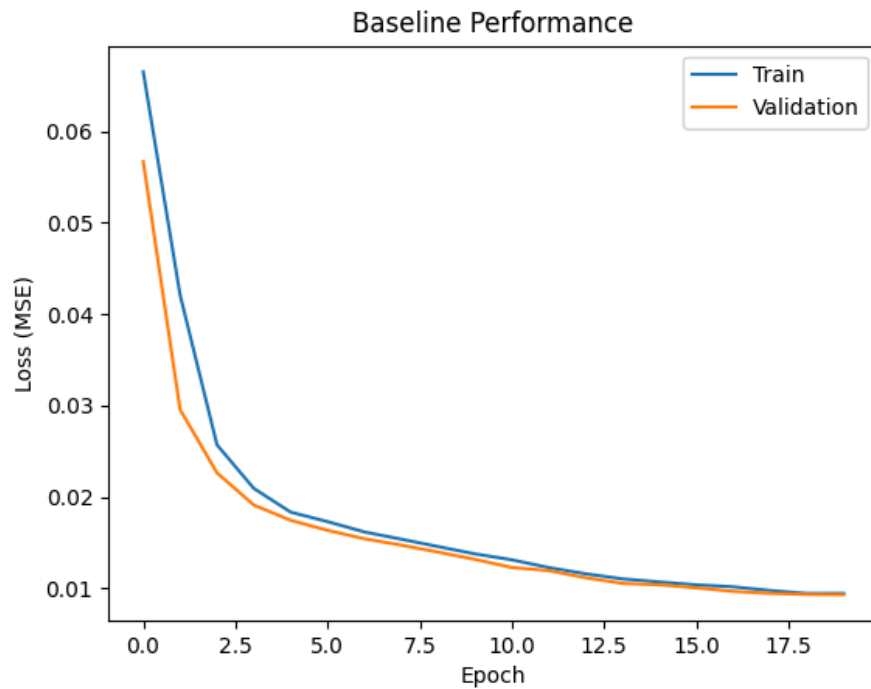


Figure 4: Loss (MSE) for baseline preprocessing

Performance after normalizing to $[-1,1]$:

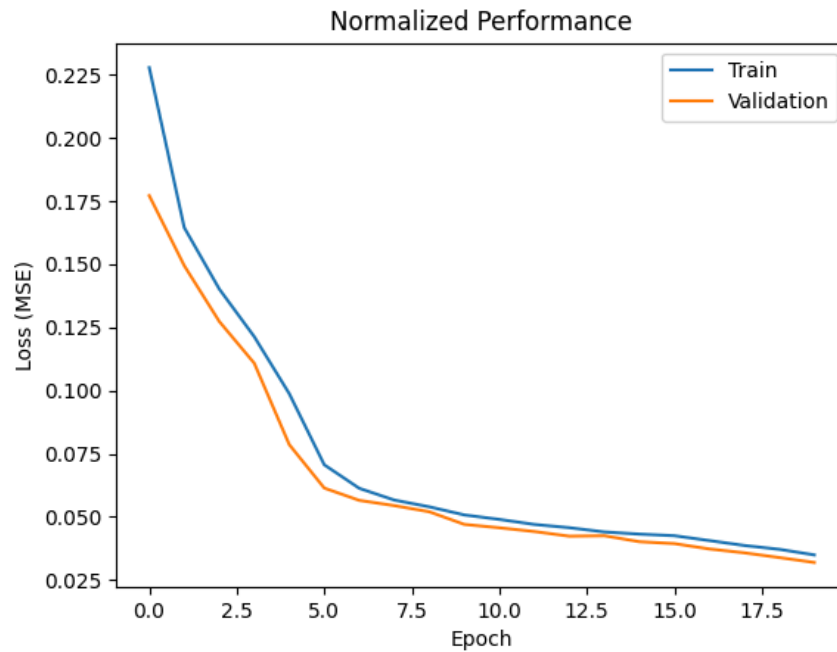


Figure 5: Loss (MSE) for normalized inputs

Performance after adding noise:

Note that noise was added by sampling values from the normal distribution, scaling them by the standard deviation, and shifting with the mean. This noise is then added to the original image (tensor).

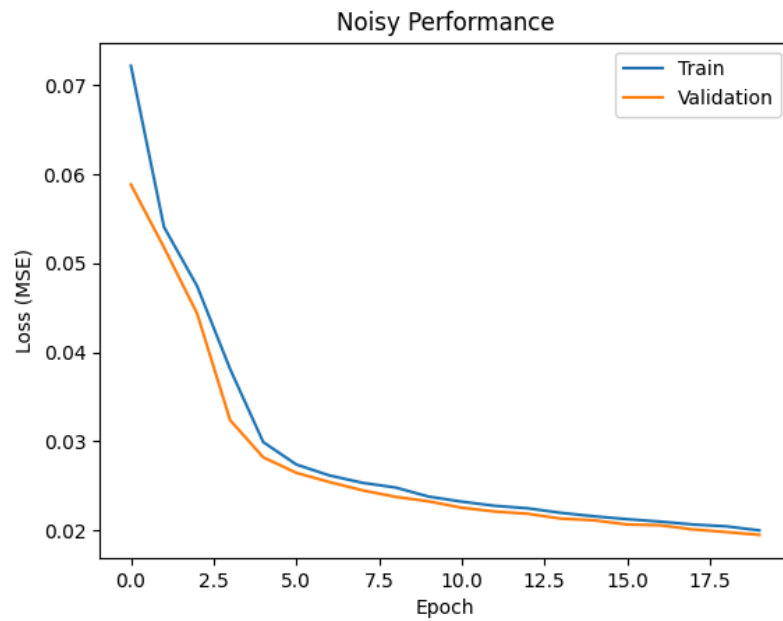


Figure 6: Loss (MSE) for noisy inputs

Performance after adding noise to the normalized $[-1, 1]$ data:

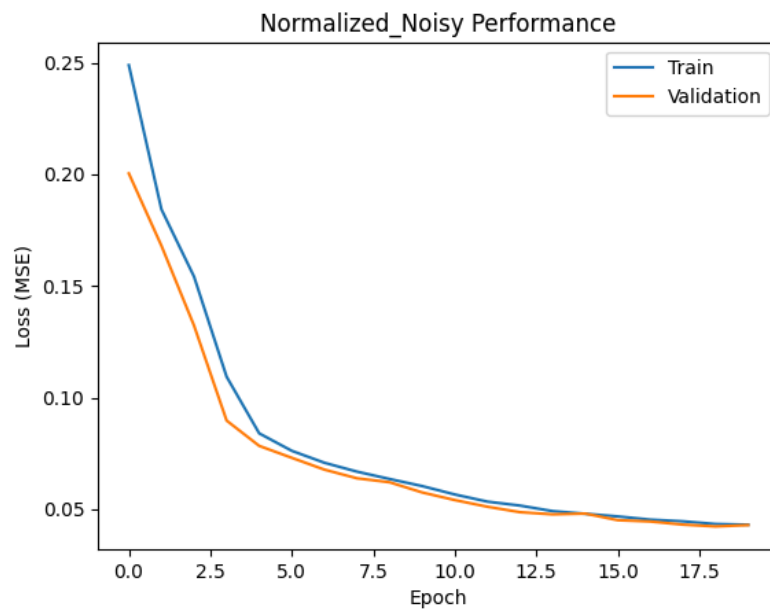


Figure 7: Loss (MSE) for noisy and normalized inputs

The performance when preprocessing with noise for both normalized and non-normalized inputs is worse than without noise. This is to be expected, as adding noise to the input makes the image reconstruction (of the original image) more difficult. However, this may generalize better to unseen data and perform better at test time.

Furthermore, we see the baseline performance converge faster and perform better compared to the noisy experiments.

b) Batch Normalization

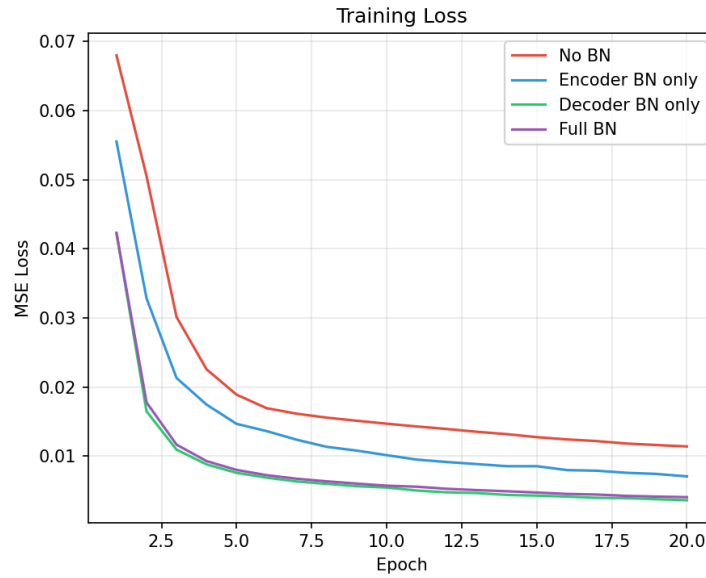


Figure 8: Training loss (MSE) for different batch normalizations

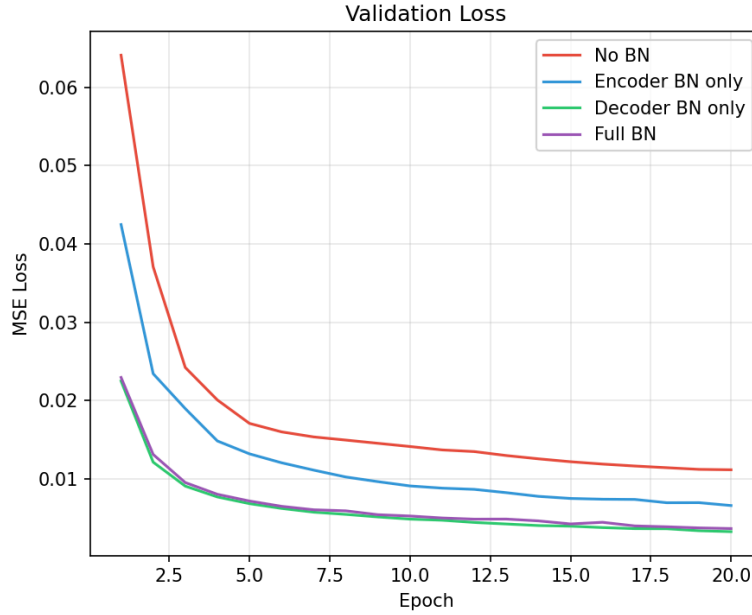


Figure 9: Validation loss (MSE) for different batch normalizations

Batch normalization (BN) consistently improves convergence speed and final MSE compared to no BN. Decoder BN has the largest impact. Both "Decoder BN only" and "Full BN" converge fastest and reach the lowest loss, while "Encoder BN only" lags behind. This suggests stabilizing activations in the reconstruction path is more important than in the encoding path for this autoencoder.

c) L1 and L2 Regularization

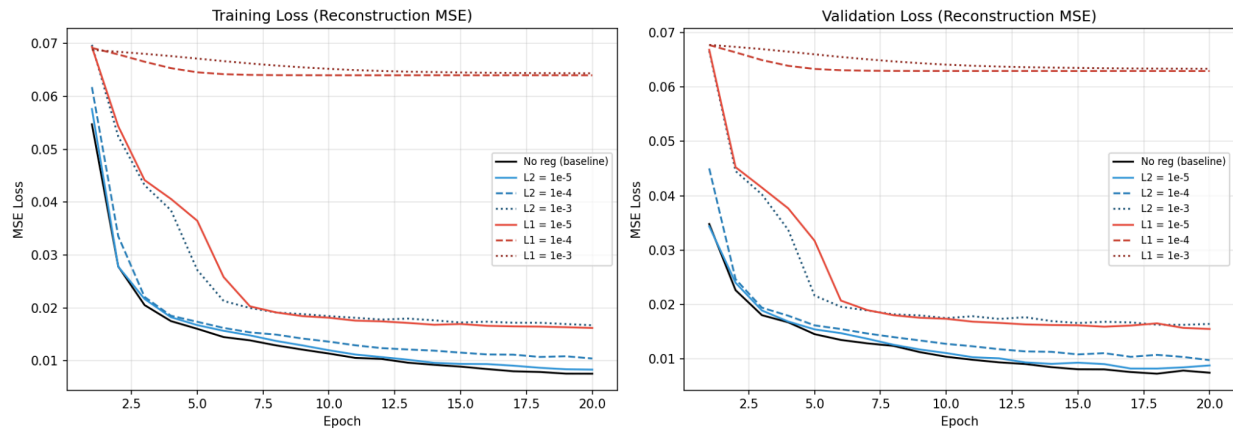


Figure 10: Training and Validation loss (MSE) for L1 and L2 regularization

We see that no regularization produces the best results. This is expected because the autoencoder was not significantly overfitting to begin with. Regularization, in this case, is worsening the results rather than improving them.

d) Dropout

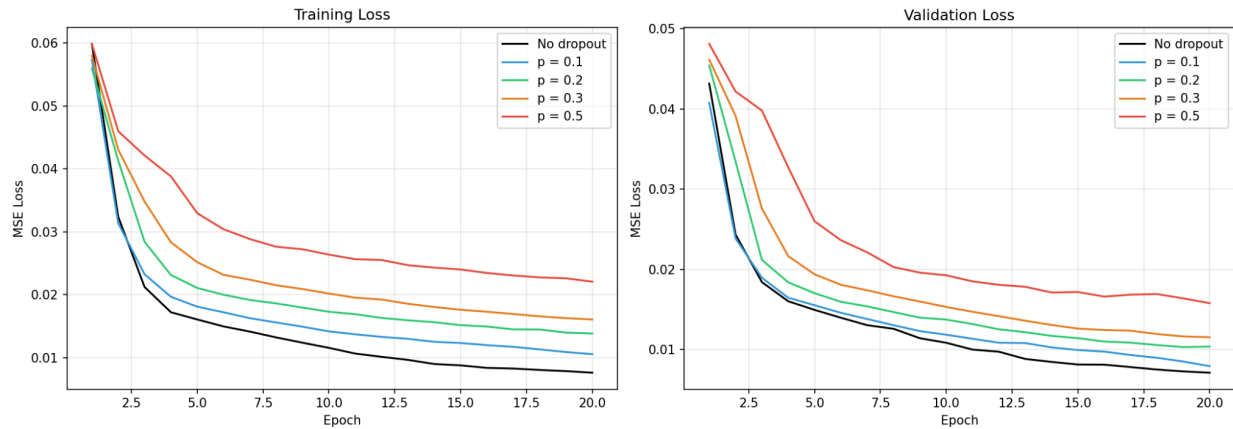


Figure 11: Training and Validation loss (MSE) for different dropouts

Higher dropout consistently increases training loss — expected, since dropout deliberately disables neurons during training, making the task harder. The ordering is: no dropout < $p=0.1$ < $p=0.2$ < $p=0.3$ < $p=0.5$.

e) Learning Rate

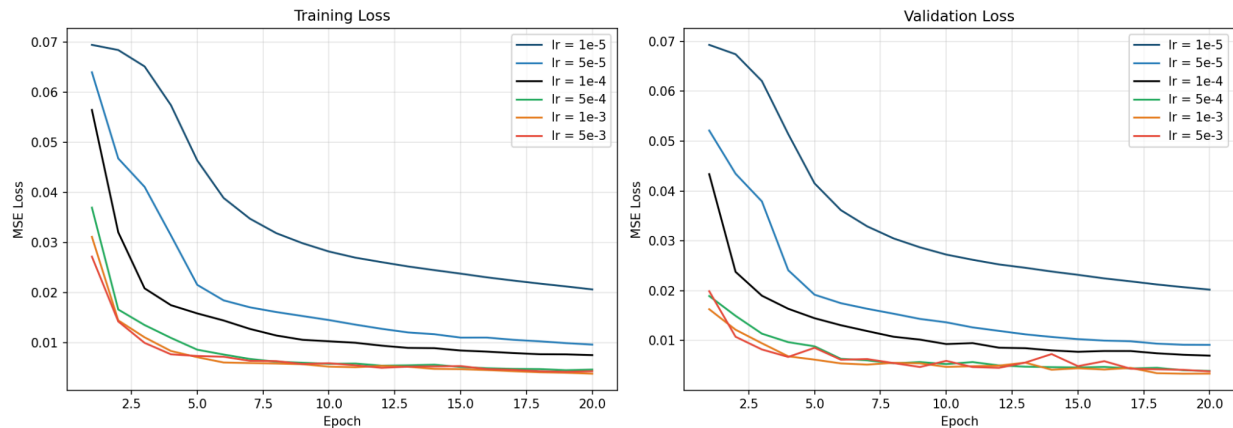


Figure 12: Training and Validation loss (MSE) for different learning rates

Lower learning rates ($1e-5$, $5e-5$) converge too slowly within 20 epochs. Higher rates ($5e-4$ to $1e-3$) converge fastest and reach the lowest loss. The optimal range is $5e-4$ to $1e-3$, faster than the baseline $1e-4$ without sacrificing stability.

f) Best Performing Model

Optimal Parameters:

- No regularization
- No dropout
- Learning rate: 0.0005
- 30 Epochs

Results (MSE Loss):

Training Set: 0.00184

Validation Set: 0.00151

Test Set: 0.00147